higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

DATA COMMUNICATION AND NETWORKING
NQF LEVEL 4

(10041014)

27 February 2019 (Y-Paper)
13:00–16:00

This question paper consists of 11 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NC(V)

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.20) in the ANSWER BOOK.

1.1 Data sharing enables groups of users to …

A exchange information.
B use one peripheral at a time.
C support electronic mail.
D restrict access to information.

1.2 A protocol suite that enables two computers to establish a connection and exchange streams of data is …

A SMTP.
B ATM.
C IEEE 802.11.
D TCP/IP.

1.3 A computer that can send and receive data over a wireless connection is known to be …

A networked.
B Wi-Fi enabled.
C a dial-up connection.
D hardwired.

1.4 Which ONE of the following is NOT a network topology?

A Star
B Bus
C Tree
D Ring

1.5 The technique in which enough of the frame is being read to guarantee that the frame is at least the minimum size for the network is called …

A store and forward switching.
B fragment free switching.
C half switching.
D packet switching.

1.6 … is Novell's protocol for packet routing and forwarding.

A TCP/IP
B IPX/SPX
C DHCP
D IPV6

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1.7 You have been hired to install a network at a large government agency that wants to reduce electronic eavesdropping on its network.

What kind of cable will you use?

A UTP
B STP
C Coaxial
D Fibre optic

1.8 A ... device does NOT need extra software for it to function as the operating system already contains its specifications.

A NIC
B printer
C plug-and-play
D switch

1.9 A program a user can run to install an application is known as …

A application software.
B setup executable file.
C read only file.
D control panel.

1.10. Which WAN device is used as an interface between a router and a digital circuit?

A WAN switch
B Modem
C ISDN terminal adapter
D CSU/DSU

1.11 Which layer in the OSI Reference model ensures that data sent by the computer can be read by the destination computer?

A Session
B Presentation
C Application
D Physical

1.12 What is the purpose of a modem?

A Converts data into electrical signals
B Converts analogue signals into digital and vice versa
C Boosts digital signals on long distance lines
D Reroutes signals when a channel is busy

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1.13 What is CSMA/CD?

A Node access method
B Wiring system
C Topology
D An ITU standard

1.14 Which connector shape is used to connect twisted pair cable to network devices?

A Barrel
B RJ
C Y
D T

1.15 Which ONE of the following will not be part of a network performance report?

A Network performance baselines
B Forecasts
C Network consumables
D Increase in network activity

1.16 What is the minimum requirement of hard disk space for a client computer that will need to run Windows XP Professional?

A 1GB
B 8GB
C 1,5 GB
D 64 MB

1.17 Which account will have unlimited rights to the network?

A Guest
B Administrator
C User
D Foreign

1.18 An event/situation that might happen in the future that could change the current plan is known as a …

A contingency plan.
B backup plan.
C monitoring.
D network security.

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1.19 The number of network devices on a route between a sending node and a receiving node is indicated by …

A router.
B switch.
C time delay.
D hop count.

1.20 What is the name of the unique address that is assigned to a network device before it leaves the factory?

A IP address
B MAC address
C Node access address
D TCP/IP address

(20 × 1) **[20]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 A data source is the computer that sends a message across a network to another computer.

2.2 Full duplex communication takes place in both directions at the same time.

2.3 A passive topology is one in which the computer and devices regenerate signals to move data through the network.

2.4 A baseline is a measurement of performance taken under what is considered a normal load.

2.5 A laser is an intense focused light.

2.6 A plug-and-play device will need extra software to function because the operating system does not contain the devices specifications.

2.7 A FAQ is a list of frequently asked questions about a particular subject.

2.8 A wizard is a utility that helps users monitor the performance of a network.

2.9 A service pack is an updated version of a software package.

2.10 A WAN covers a large geographical area and is used to link LANs in different cities or towns.

(10 × 1) **[10]**

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QUESTION 3

Complete the following sentences by filling in the missing word(s). Write only the word(s) next to the question number (3.1–3.10) in the ANSWER BOOK.

3.1 A … is a unit of information that can be transmitted over a computer network.

3.2 Fibre optic cables have a … or plastic inner core that carry data signals.

3.3 Each NIC has a unique identifier called a … address.

3.4 … is a network layer protocol that provides source and destination addressing and routing for the TCP/IP suite.

3.5 The token ring network architecture can isolate faults automatically by using a process called …

3.6 … is a process that involves finding out if it is possible to meet the user's requirements, considering budget and time.

3.7 Programmes that help users configure, customise and work with their computer and network environment is called …

3.8 You can draw conclusions about server activities by observing details of …

3.9 … is a document that describes the rules governing access to a company's information resources.

3.10 Active hubs have a/an … that can monitor network activity and store monitoring information.

(10 × 1) **[10]**

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QUESTION 4

Choose an item from COLUMN B that matches the description in COLUMN A. Write only the letter (A–M) next to the question number (4.1–4.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.1	Process of looking for problems, diagnosing them and correcting them	A	access control
4.2	A network topology in which all computers connect to each other	B	full-duplex
4.3	Each computer in the network can be either a client or a server	C	modem
4.4	Handles error-free delivery and controls the placement of frames on the media	D	diagnostic software
4.5	Rules determining whether a packet should be permitted to reach the intended destination	E	domain controller
4.6	Presents data in the correct format	F	troubleshooting
4.7	Specialised programmes that can probe and monitor a system	G	data link layer
4.8	Special purpose network device that allows a computer to send and receive network signals over a landline connection	H	mesh
4.9	Data transmission take place in both directions at the same time	I	presentation layer
4.10	A directory server that provides access controls over users, accounts, groups and other networks	J	peer-to-peer network
		K	routing table
		L	signal bounce
		M	ring network

(10 × 1) **[10]**

TOTAL SECTION A: 50

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SECTION B

QUESTION 5

5.1 Identify the types of LAN media depicted below.



A



B (2)

5.2 Describe the steps needed to configure a static IP address in Windows. (8)



FIGURE 1

5.3 Give a brief description of the concept *depicted* in FIGURE 1. (4)

5.4 Specify the SIX basic components of a telecommunications network. (6)
[20]

QUESTION 6

6.1 Study the IP address given below and answer the questions.

174.24.208.192

6.1.1 Determine the network address. (1)

6.1.2 Determine the host ID. (1)

6.1.3 Formulate an IP address for another PC on the same network. (2)

6.2 By means of a labelled diagram, illustrate the components needed for a traditional LAN and how they are connected. (6)

6.3 Differentiate between a *public key encryption* and a *private key encryption*. (4)

6.4 Explain how a client/server network configuration works. (6)
[20]

QUESTION 7

7.1 Describe how to go about making a Cat 5/5e or Cat 6 UTP crossover cable. (6)

7.2 Name FOUR tools and materials needed to complete making the cable in QUESTION 7.1. (4)

Read the scenario below and answer the questions.

Mr Zungu owns Clearview Optics. His company supplies sunglasses, reading glasses and tested glasses to the public. He has FIVE stores within the Durban area. To make his business more efficient he has decided to network his FIVE stores.

7.3 Name SIX tasks that he will have to include in his LAN installation plan. (6)

7.4 List FOUR ways in which you will equip Mr Zungu's staff with the knowledge and skills they need to take advantage of the network's capabilities. (4)
[20]

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QUESTION 8

8.1 Explain the concept of a *network application.* (2)

8.2 You have been asked to install a network operating system.

List EIGHT steps to consider when preparing to install a network operating system. (8)

8.3 Analyse how a print queue is managed. (5)

8.4 Identify THREE functions of built-in utilities. (3)

8.5 Name TWO ways of remotely accessing a network. (2)
[20]

QUESTION 9

9.1 List SEVEN steps to follow in order to shut down a network properly. (7)

9.2 The network is available, but a user is unable to access it.

Name THREE possible reasons why the user cannot access the network. (3)

9.3 Explain how a packet switching network handles data. (5)

9.4 Distinguish between a *switched virtual circuit* and *permanent virtual circuit*. (5)
[20]

TOTAL SECTION B: 100
GRAND TOTAL: 150

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